

sky130_rhbd_tt_1P8_25C.ccs Library

Cell Groups
AND2X1
AND3X1
AO3X1
AOA4X1
AOAI4X1
AOI3X1
BUFX1
DFFQNX1
DFFQX1
DFFRNQNX1
DFFRNQX1
DFFRNX1
DFFSNQNX1
DFFSNQX1
DFFSNRNQNX1
DFFSNRNQX1
DFFSNRNX1
DFFSNX1
DFFX1
DLATCHN
DLATCH
FA
HA

INVX1
INV
MUX2X1
NAND2X1
NAND3X1
NOR2X1
NOR3X1
OR2X1
OR3X1
TIEHI
TIELO
TMRDFFQNX1
TMRDFFQX1
TMRDFFRNQX1
TMRDFFSNQNX1
TMRDFFSNQX1
TMRDFFSNRNQNX1
TMRDFFSNRNQX1
VOTER3X1
VOTERN3X1
XNOR2X1
XOR2X1

AND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
AND2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
AND2X1	0.01040	0.01053	6.00091

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND2X1	0.00000	7.37298	13.28970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (RR)	0.05424	0.10868	0.03666
	B->Y (RR)	0.04789	0.11360	0.14631

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (FF)	0.04661	0.15469	1.14368
	B->Y (FF)	0.04386	0.13511	0.66099

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	8.25087	8.25704	8.53729
	B	0.00000	0.00000	0.00000
	B	8.25188	8.26000	8.55527

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	2.39226	2.40429	2.75200
	B	0.00000	0.00000	0.00000
	B	2.38413	2.39302	2.65576

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.55827	2.55789	2.55761

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.15085	3.14997	3.14991

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.56143	2.56105	2.56098

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.15104	3.15112	3.15158

AND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	0
1	0	x	0
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AND3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AND3X1	0.01048	0.01028	0.01052	6.07005

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND3X1	0.00000	6.97639	20.76320

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (RR)	0.07980	0.13481	0.18904
	B->Y (RR)	0.07337	0.14153	0.28676
	C->Y (RR)	0.06500	0.14759	0.41869

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (FF)	0.05314	0.16878	1.21795
	B->Y (FF)	0.05099	0.15569	0.81663
	C->Y (FF)	0.04698	0.13856	0.48298

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	13.03930	13.04270	13.30840
	B	0.00000	0.00000	0.00000
	B	13.03970	13.04410	13.30990
	C	0.00000	0.00000	0.00000
	C	13.04030	13.04620	13.32170

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	1.66317	1.67483	2.04157
	B	0.00000	0.00000	0.00000
	B	1.65430	1.66376	1.94631
	C	0.00000	0.00000	0.00000
	C	1.64602	1.65343	1.88685

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.22192	2.22146	2.22118
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.22424	2.22379	2.22358
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22057	2.22031	2.22024

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.78098	2.77964	2.77988
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.78229	2.78135	2.78120
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.77925	2.77881	2.77896

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.22215	2.22191	2.22180
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22744	2.22717	2.22654
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.22084	2.21916	2.22057

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.78056	2.77946	2.77941
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.78221	2.78203	2.78225
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.78036	2.78022	2.78056

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.22514	2.22446	2.22459
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.22832	2.22731	2.22769

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.78114	2.78097	2.78109
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.78069	2.78018	2.78034

AO3X1

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Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AO3X1	76.51360

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AO3X1	0.01029	0.01056	0.01041	5.69722

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AO3X1	0.00000	5.28280	19.23130

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (FR)	0.08795	0.20696	1.03473
	B->Y (FR)	0.08410	0.18865	0.69345
	C->Y (RR)	0.03740	0.08715	-0.32576

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (RF)	0.12667	0.18903	0.07691
	B->Y (RF)	0.11926	0.19178	0.17900
	C->Y (FF)	0.06448	0.16942	1.01297

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	1.49677	1.50541	1.78599
	B	0.00000	0.00000	0.00000
	B	1.48732	1.49458	1.73441
	C	0.00000	0.00000	0.00000
	C	11.37850	11.38340	11.61440

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	10.20550	10.20540	10.41560
	B	0.00000	0.00000	0.00000
	B	10.20810	10.20910	10.43290
	C	0.00000	0.00000	0.00000
	C	11.39470	11.40210	11.63450

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	11.81750	11.82050	12.06010
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.03477	0.03474	0.03471

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06954	0.07715	0.34459
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.05856	0.05801	0.05787

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	11.81740	11.82250	12.06570
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.03741	0.03734	0.03734

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06092	0.06731	0.29916
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.05814	0.05832	0.05828

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * !B * Y) + (!A * Y)	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	0.02397	0.02394	0.02393

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	$(A * !B * Y) + (!A * Y)$	0.00000	0.00000	0.00000
	$(A * !B * Y) + (!A * Y)$	1.47391	1.47388	1.47397

AOA4X1

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1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
A	B	C	D	Y
0	x	x	x	0
1	0	x	x	0
1	1	x	0	0
1	1	0	1	1
1	1	1	1	0

Footprint

Cell Name	Area
AOA4X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	Y
AOA4X1	0.01024	0.01052	0.01048	0.01044	5.51490

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOA4X1	0.00000	10.36610	25.24350

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (RR)	0.18142	0.24313	0.10862
	B->Y (RR)	0.17380	0.24557	0.21384
	C->Y (FR)	0.11801	0.23279	1.14092
	D->Y (RR)	0.04995	0.11747	0.07090

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (FF)	0.12801	0.24676	1.01911
	B->Y (FF)	0.12421	0.22915	0.72855
	C->Y (RF)	0.07695	0.13618	-0.28053
	D->Y (FF)	0.04474	0.13674	0.58729

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	14.30690	14.30790	14.52130
	B	0.00000	0.00000	0.00000
	B	14.30900	14.30980	14.53830
	C	0.00000	0.00000	0.00000
	C	15.16170	15.16780	15.37720
	D	0.00000	0.00000	0.00000
	D	15.24170	15.24770	15.50750

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	4.88096	4.88849	5.17158
	B	0.00000	0.00000	0.00000
	B	4.87258	4.87857	5.13064
	C	0.00000	0.00000	0.00000
	C	14.49990	14.50070	14.71920
	D	0.00000	0.00000	0.00000
	D	9.30796	9.31333	9.56116

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	15.38050	15.38420	15.61830
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	8.11210	8.11379	8.30200
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.75254	2.75236	2.75269

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	2.51133	2.51867	2.77991
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	5.02328	5.03048	5.27783
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.06300	3.06250	3.06239

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	15.38110	15.38530	15.62310
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	8.11959	8.11888	8.31390
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.75364	2.75399	2.75437

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * C * !Y)$	0.00000	0.00000	0.00000
	$(A * C * !Y)$	2.50483	2.51031	2.74504
	$(A * !C * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * !C * !D * !Y)$	5.01551	5.02171	5.24480
	$(!A * !Y)$	0.00000	0.00000	0.00000
	$(!A * !Y)$	3.06282	3.06289	3.06311

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * B * !D * !Y)$	15.21140	15.21640	15.42700
	$(A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * !B * !Y) + (!A * !Y)$	1.97980	1.97963	1.97966

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * B * !D * !Y)$	9.30785	9.31351	9.51923
	$(A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * !B * !Y) + (!A * !Y)$	5.01110	5.01027	5.01020

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	14.46460	14.46280	14.46420

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	15.10390	15.10380	15.10390

AOAI4X1

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Voltage 1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
A	B	C	D	YN
0	x	x	x	1
1	0	x	x	1
1	1	x	0	1
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
AOAI4X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	YN
AOAI4X1	0.01026	0.01052	0.01048	0.01050	5.27895

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOAI4X1	0.00000	5.05158	26.53990

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (FR)	0.09925	0.21712	0.98165
	B->YN (FR)	0.09477	0.19862	0.67585
	C->YN (RR)	0.04774	0.10524	-0.29584
	D->YN (FR)	0.01639	0.06789	0.40490

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (RF)	0.14355	0.20413	0.04505
	B->YN (RF)	0.13587	0.20575	0.13757
	C->YN (FF)	0.08062	0.19231	1.10437
	D->YN (RF)	0.01798	0.04207	-0.19000

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	1.33483	1.34246	1.64294
	B	0.00000	0.00000	0.00000
	B	1.32655	1.33319	1.59291
	C	0.00000	0.00000	0.00000
	C	9.89351	9.89380	10.10730
	D	0.00000	0.00000	0.00000
	D	5.16879	5.17515	5.40784

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	15.44000	15.44070	15.66030
	B	0.00000	0.00000	0.00000
	B	15.44290	15.44230	15.67350
	C	0.00000	0.00000	0.00000
	C	16.30600	16.31330	16.52120
	D	0.00000	0.00000	0.00000
	D	16.33020	16.33800	16.58520

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	10.88330	10.88650	11.11590
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	4.22506	4.22370	4.40557
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.03323	0.03322	0.03318

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	0.07092	0.07886	0.35181
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	1.41086	1.41840	1.66445
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.05478	0.05422	0.05409

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	10.88320	10.88740	11.12360
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	4.22834	4.22717	4.41882
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.03526	0.03520	0.03524

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	0.06287	0.06961	0.30869
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	1.40291	1.40929	1.63084
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.05446	0.05457	0.05460

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	10.82310	10.82640	11.03900
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	0.02648	0.02646	0.02645

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	5.42402	5.42897	5.64020
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	1.37274	1.37282	1.37285

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.13040	10.12790	10.13020

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	$(A * B * C * YN) + (A * !B * YN) + (!A * YN)$	0.00000	0.00000	0.00000
	$(A * B * C * YN) + (A * !B * YN) + (!A * YN)$	10.75610	10.75580	10.75620

AOI3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	x	x	0
1	0	x	0
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
AOI3X1	59.28640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
AOI3X1	0.01037	0.01062	0.01049	2.87215

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOI3X1	0.00000	4.61712	20.74590

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (RR)	0.08047	0.14141	0.05960
	B->YN (RR)	0.07453	0.14769	0.17964
	C->YN (FR)	0.02207	0.07945	0.61637

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (FF)	0.05806	0.17394	1.08104
	B->YN (FF)	0.05476	0.15538	0.68676
	C->YN (RF)	0.01266	0.01576	-0.57178

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	5.57877	5.57698	5.78114
	B	0.00000	0.00000	0.00000
	B	5.57963	5.58159	5.79398
	C	0.00000	0.00000	0.00000
	C	6.78365	6.78973	6.99673

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	1.63372	1.64335	1.93211
	B	0.00000	0.00000	0.00000
	B	1.62556	1.63274	1.87048
	C	0.00000	0.00000	0.00000
	C	12.73790	12.74530	12.97230

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	13.00820	13.01230	13.27220
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.03501	0.03499	0.03496

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.06376	0.07051	0.32219
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.06103	0.06048	0.06034

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	13.00710	13.01290	13.27270
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.03699	0.03693	0.03691

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.05557	0.06116	0.27512
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.06090	0.06109	0.06105

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	0.01823	0.01820	0.01818

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	1.60724	1.60668	1.60686

BUFX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT
A
x

Footprint

Cell Name	Area
BUFX1	42.05920

Pin Capacitance Information

Cell Name	Pin Cap(pf)
	A
BUFX1	0.01028

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
BUFX1	0.00000	6.07080	6.07080

Passive Power Information

Passive power(pJ) for A rising :

Cell Name	Power(pJ)		
	first	mid	last
BUFX1	0.00000	0.00000	0.00000
	3.52075	3.53123	3.84177

Passive power(pJ) for A falling :

Cell Name	Power(pJ)		
	first	mid	last
BUFX1	0.00000	0.00000	0.00000
	3.54276	3.55412	3.86609

DFFQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
DFFQNX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
DFFQNX1	0.01055	0.02318	5.42591

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQNX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (RR)	0.14049	0.22895	0.34019

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (RF)	0.15937	0.27222	0.40546

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	0.04879	0.07572	0.80329
	setup	CLK (R)	0.10095	0.13658	0.76827

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	-0.03661	-0.06975	-0.73997
	setup	CLK (R)	0.06264	0.10469	1.59870

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.10340	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.13112	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.18657	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.09416	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25720	26.25600	26.47800

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13040	21.13130	21.36170

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	24.93700	24.93590	24.93790
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	21.09480	21.09540	21.31620
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29660	11.29880	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	25.56830	25.56880	25.56920
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	10.10460	10.10790	10.34910
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30910	11.31320	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	21.09780	21.09630	21.30850
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	26.21620	26.21560	26.41830

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	10.32760	10.32800	10.54020
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	10.11700	10.11990	10.35220
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	9.90476	9.90815	10.11400
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	11.32040	11.32520	11.55020

DFFQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	R	0
1	R	1
x	x	IQ

Footprint

Cell Name	Area
DFFQX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
DFFQX1	0.01057	0.02318	5.42070

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (RR)	0.11320	0.22474	0.33781

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (RF)	0.19118	0.28023	0.40058

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	0.04564	0.07077	0.73940
	setup	CLK (R)	0.10732	0.15037	0.81645

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	-0.03337	-0.06975	-0.73888
	setup	CLK (R)	0.05919	0.09965	1.54439

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.10340	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.13112	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.19273	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.09108	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13120	21.13070	21.36150

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25710	26.25600	26.47720

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	21.09460	21.09520	21.31600
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	24.93710	24.93590	24.93790
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29400	11.29580	11.50830

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	10.10460	10.10790	10.34910
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	25.56820	25.56880	25.56920
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30790	11.31160	11.53250

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	21.09760	21.09580	21.30850
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	26.21450	26.21510	26.41790

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	10.11920	10.12200	10.35450
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	10.32750	10.32770	10.54000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	11.31990	11.32380	11.54880
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.90787	9.90992	10.11610

DFFRNQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	RN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	1
x	1	x	IQN

Footprint

Cell Name	Area
DFFRNQNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	QN
DFFRNQNX1	0.01013	0.03073	0.02340	3.68547

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RR)	0.15328	0.24222	0.34019
	RN->QN (FR)	0.03369	0.13916	0.78163

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RF)	0.22551	0.32650	0.58283

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	0.04104	0.05830	0.86996
	setup	CLK (R)	0.13588	0.15927	0.61047

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	-0.02038	-0.05979	-0.69270
	setup	CLK (R)	0.07017	0.11354	1.92432

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	0.04104	0.05830	0.86996
	setup	CLK (R)	RN	0.13588	0.15927	0.61047

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	-0.02038	-0.05979	-0.69270
	setup	CLK (R)	RN	0.07017	0.11354	1.92432

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	recovery	CLK (R)	0.12509	0.16750	7.07569
	removal	CLK (R)	0.03386	0.03986	0.01946

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	recovery	CLK (R)	D	0.12509	0.16750	7.07569
	removal	CLK (R)	D	0.03386	0.03986	0.01946

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	RN ()	(CLK * D)	0.11880	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.08800	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.05411	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.05411	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.15884	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.22045	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10340	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78440	23.78230	23.99460
	RN	24.23140	24.23900	24.60350

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51670	34.51450	34.72350

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	22.75360	22.75650	22.75800
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	34.32650	34.32680	34.52350
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	22.73540	22.74100	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	18.84300	18.84340	19.03820
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	9.85878	9.85875	9.85928

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	23.36770	23.36940	23.37000
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	16.53740	16.54090	16.76100
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	23.35810	23.36020	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	13.88420	13.88840	14.10360
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	10.43060	10.42950	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	22.01870	22.01610	22.01630
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	22.01030	22.00780	22.00910
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	13.81990	13.81880	14.02080
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	9.22431	9.22530	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	23.85180	23.85380	23.85450
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	23.85320	23.85460	23.85550
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	10.54300	10.54590	10.74640
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	10.91510	10.91390	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	34.32910	34.32590	34.51390
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	23.89470	23.89400	24.08350
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	23.90760	23.90700	24.10180
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	23.89240	23.89150	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * QN)	0.00000	0.00000	0.00000
	(D * RN * QN)	13.81060	13.81130	13.99940
	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	16.55680	16.55460	16.77150
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	9.21269	9.21213	9.40994
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	9.21874	9.21970	9.42132
	(!D * RN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !QN)	13.90130	13.90590	14.13170
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	9.21432	9.21812	9.41697

DFFRNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process
, Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	0
x	1	x	IQ

Footprint

Cell Name	Area
DFFRNQX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
DFFRNQX1	0.01013	0.03072	0.02340	5.48149

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RR)	0.16114	0.26114	0.49468

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RF)	0.20452	0.29367	0.39820
	RN->Q (FF)	0.11068	0.25526	0.97656

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	0.03723	0.05198	0.79082
	setup	CLK (R)	0.14389	0.17818	0.64824

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	-0.01972	-0.05979	-0.69341
	setup	CLK (R)	0.06185	0.10640	1.85725

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	0.03723	0.05198	0.79082
	setup	CLK (R)	RN	0.14389	0.17818	0.64824

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	-0.01972	-0.05979	-0.69341
	setup	CLK (R)	RN	0.06185	0.10640	1.85725

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	recovery	CLK (R)	0.13259	0.18297	7.32559
	removal	CLK (R)	0.03386	0.03986	0.01946

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	recovery	CLK (R)	D	0.13259	0.18297	7.32559
	removal	CLK (R)	D	0.03386	0.03986	0.01946

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	RN ()	(CLK * D)	0.11880	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.08800	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.05411	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.05411	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.15884	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.22661	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10032	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51590	34.51480	34.72320

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78450	23.78310	23.99360
	RN	-0.00000	-0.00000	-0.00000
	RN	24.23210	24.23850	24.60970

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	34.32650	34.32680	34.52350
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	22.75360	22.75650	22.75800
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	22.73580	22.74100	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	14.42910	14.43040	14.61840
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	9.85877	9.85875	9.85927

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	16.53740	16.54090	16.76090
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	23.36770	23.36940	23.37000
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	23.35810	23.36030	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	9.99547	9.99947	10.21160
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	10.43060	10.42950	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	22.01870	22.01610	22.01630
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	22.01030	22.00780	22.00910
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	13.81990	13.81880	14.02080
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	9.22425	9.22530	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	23.85190	23.85380	23.85450
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	23.85330	23.85460	23.85550
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	10.54310	10.54590	10.74610
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	10.91520	10.91390	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	34.32910	34.32590	34.51390
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	23.89460	23.89400	24.08350
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	23.90760	23.90700	24.10180
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	23.89240	23.89150	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	16.55680	16.55460	16.77150
	(D * RN * !Q)	0.00000	0.00000	0.00000
	(D * RN * !Q)	13.81070	13.81130	13.99940
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	9.21269	9.21222	9.40995
	(!D * RN * Q)	0.00000	0.00000	0.00000
	(!D * RN * Q)	13.90130	13.90590	14.13170
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	9.21874	9.21974	9.42132
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	9.21431	9.21811	9.41697

DFFRNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT	
D	RN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	0	1
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFRNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	RN	CLK	Q	QN
DFFRNX1	0.01013	0.03073	0.02340	5.44593	3.68531

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RR)	0.16119	0.26114	0.49469
	QN->Q (FR)	0.03595	0.13386	0.78119

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RF)	0.20444	0.29359	0.39812
	QN->Q (RF)	0.03980	0.07488	-0.28507
	RN->Q (FF)	0.11078	0.25535	0.97662

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RR)	0.15328	0.24222	0.34019
	Q->QN (FR)	0.03292	0.11921	0.38902
	RN->QN (FR)	0.03372	0.13919	0.78164

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RF)	0.22544	0.32643	0.58275
	Q->QN (RF)	0.04844	0.11628	0.05458

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	0.03723	0.05199	0.79092
	setup	CLK (R)	0.14385	0.17817	0.64826

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	-0.01972	-0.05979	-0.69341
	setup	CLK (R)	0.06186	0.10641	1.85744

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	0.03723	0.05199	0.79092
	setup	CLK (R)	RN	0.14385	0.17817	0.64826

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	-0.01972	-0.05979	-0.69341
	setup	CLK (R)	RN	0.06186	0.10641	1.85744

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	recovery	CLK (R)	0.13391	0.18606	7.33515
	removal	CLK (R)	0.03386	0.03986	0.01946

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	recovery	CLK (R)	D	0.13391	0.18606	7.33515
	removal	CLK (R)	D	0.03386	0.03986	0.01946

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	RN ()	(CLK * D)	0.11880	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.08800	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.05411	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.05411	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.15884	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.22661	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10032	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25800	17.25740	17.36160

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89220	11.89130	11.99680
	RN	-0.00000	-0.00000	-0.00000
	RN	12.11590	12.11930	12.30490

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89220	11.89120	11.99720
	RN	-0.00000	-0.00000	-0.00000
	RN	12.11580	12.11960	12.30170

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25840	17.25740	17.36170

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	34.32640	34.32680	34.52350
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	22.75360	22.75650	22.75800
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	22.73540	22.74100	22.74400
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	18.84300	18.84340	19.03820
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	9.85878	9.85875	9.85928

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	16.53740	16.54080	16.76090
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	23.36770	23.36940	23.37000
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	23.35810	23.36020	23.36120
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	13.88410	13.88830	14.10360
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	10.43060	10.42950	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	22.01830	22.01610	22.01630
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	22.00920	22.00790	22.00910
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	13.81930	13.81900	14.02080
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	9.22374	9.22531	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	23.85340	23.85380	23.85450
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	23.85340	23.85460	23.85550
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	10.54360	10.54590	10.74640
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	10.91480	10.91390	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	34.32910	34.32590	34.51390
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	23.89460	23.89400	24.08350
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	23.90760	23.90700	24.10180
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	23.89240	23.89150	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	16.55710	16.55490	16.77180
	(D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * !Q * QN)	13.81070	13.81130	13.99940
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	9.21268	9.21212	9.40994
	(!D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * Q * !QN)	13.90130	13.90580	14.13170
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	9.21876	9.21971	9.42133
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	9.21440	9.21803	9.41697

DFFSNQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	0
x	1	x	IQN

Footprint

Cell Name	Area
DFFSNQNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
DFFSNQNX1	0.01015	0.02175	0.02229	5.23349

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RR)	0.13837	0.22461	0.32665

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RF)	0.15612	0.26219	0.32646
	SN->QN (FF)	0.24161	0.35833	1.00371

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	0.01319	0.01993	0.93411
	setup	CLK (R)	0.12334	0.16034	2.10717

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	-0.06793	-0.11958	-1.15000
	setup	CLK (R)	0.10512	0.16344	1.67373

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	0.01319	0.01993	0.93411
	setup	CLK (R)	SN	0.12334	0.16034	2.10717

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	-0.06793	-0.11958	-1.15000
	setup	CLK (R)	SN	0.10512	0.16344	1.67373

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	recovery	CLK (R)	0.04108	0.03986	0.32870
	removal	CLK (R)	-0.01681	-0.01495	-0.12362

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	recovery	CLK (R)	!D	0.04108	0.03986	0.32870
	removal	CLK (R)	!D	-0.01681	-0.01495	-0.12362

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	SN ()	(CLK * D)	0.06951	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.06951	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.06027	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06027	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.10032	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13420	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.19889	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.11264	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36320	35.36260	35.58830

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64100	19.64070	19.87610
	SN	-0.00000	-0.00000	-0.00000
	SN	19.00030	19.00690	19.37000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	33.71140	33.71230	33.71360
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	19.80300	19.80250	20.00790
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	19.79740	19.79890	20.00590
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	10.28750	10.28940	10.49350

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	34.35860	34.35990	34.36010
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	9.55580	9.55812	9.77692
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	9.55979	9.56254	9.78785
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.91870	14.92430	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	18.32600	18.32720	18.33270
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	9.86855	9.87357	9.87298
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	9.87764	9.87466	9.87852
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	14.41100	14.41060	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	19.54960	19.55500	19.55620
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	11.02450	11.02740	11.02780
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	11.02790	11.02710	11.02850
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	3.67796	3.68299	3.89217

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	19.79680	19.79640	20.01980
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	19.79250	19.79200	20.01260
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	35.14020	35.13970	35.34240
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	10.63750	10.64340	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * QN)	0.00000	0.00000	0.00000
	(D * SN * QN)	13.00390	13.00240	13.24300
	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	9.56929	9.57458	9.85093
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	9.55739	9.56648	9.81653
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	17.39320	17.39360	17.60760
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	14.92490	14.92710	15.17980
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	3.48914	3.49425	3.80064

DFFSNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
DFFSNQX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
DFFSNQX1	0.01015	0.02175	0.02229	3.66982

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RR)	0.10908	0.21376	0.25769
	SN->Q (FR)	0.03581	0.14612	0.81257

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RF)	0.21366	0.30026	0.40893

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	0.01214	0.01593	0.82103
	setup	CLK (R)	0.13179	0.17384	4.28079

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	-0.06793	-0.11958	-1.15274
	setup	CLK (R)	0.10203	0.15744	1.62267

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	0.01214	0.01593	0.82103
	setup	CLK (R)	SN	0.13179	0.17384	4.28079

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	-0.06793	-0.11958	-1.15274
	setup	CLK (R)	SN	0.10203	0.15744	1.62267

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	recovery	CLK (R)	0.03351	0.03586	4.46601
	removal	CLK (R)	-0.01681	-0.01495	-0.12337

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	recovery	CLK (R)	!D	0.03351	0.03586	4.46601
	removal	CLK (R)	!D	-0.01681	-0.01495	-0.12337

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.06643	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.06643	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.06027	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06027	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.10032	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13112	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.20505	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10648	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64120	19.64120	19.87640
	SN	18.99880	19.00800	19.35870

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36330	35.36260	35.58820

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q) + (!CLK * !SN * Q)$	19.80290	19.80240	20.00770
	$(CLK * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q)$	33.71140	33.71240	33.71360
	$(CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q)$	19.79770	19.79890	20.00590
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	14.27360	14.27620	14.47990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q) + (!CLK * !SN * Q)$	9.55571	9.55812	9.77692
	$(CLK * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q)$	34.35860	34.36000	34.36010
	$(CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q)$	9.55978	9.56254	9.78784
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	19.42350	19.42850	19.64740

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * D * Q)$	18.32590	18.32710	18.33270
	$(CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(CLK * !D * Q)$	9.86839	9.87356	9.87298
	$(!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * Q)$	9.87766	9.87465	9.87851
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	14.41100	14.41030	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	19.54960	19.55500	19.55620
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	11.02450	11.02740	11.02780
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	11.02810	11.02710	11.02850
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	3.67733	3.68238	3.89154

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	19.79680	19.79640	20.01980
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	19.79250	19.79200	20.01260
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	35.14030	35.13990	35.34260
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	10.63750	10.64340	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	9.56929	9.57458	9.85092
	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	13.00400	13.00240	13.24300
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	9.55739	9.56647	9.81652
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	14.92490	14.92710	15.17980
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	17.39240	17.39360	17.60760
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	3.48913	3.49424	3.80064

DFFSNRNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80, Temp
25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	QN
0	1	1	R	1
1	1	1	R	0
x	0	x	x	1
x	1	0	x	0
x	1	1	x	IQN

Footprint

Cell Name	Area
DFFSNRNQNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	QN
DFFSNRNQNX1	0.01023	0.03416	0.02236	0.02283	3.60165

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQNX1	0.00000	25.51790	54.40870

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQNX1	CLK->QN (RR)	0.14956	0.23694	0.41752
	RN->QN (FR)	0.03425	0.14046	0.81899

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQNX1	CLK->QN (RF)	0.20143	0.29285	0.42573
	RN->QN (RF)	0.22667	0.33914	0.66800
	SN->QN (FF)	0.09644	0.21247	0.79531

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	hold	CLK (R)	-0.00976	-0.00798	2.17206
	setup	CLK (R)	0.12880	0.14221	0.36919

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.00976	-0.00798	2.17206
	setup	CLK (R)	(RN * SN)	0.12880	0.14221	0.36919

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	0.12517	0.15544	7.26260
	removal	CLK (R)	-0.01430	-0.01993	-0.06037
	hold	SN (R)	0.02299	0.04733	0.41944
	setup	SN (R)	0.02195	0.03389	1.19135

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	recovery	CLK (R)	(D * SN)	0.12517	0.15544	7.26260
	removal	CLK (R)	(D * SN)	-0.01430	-0.01993	-0.06037
	hold	SN (R)	(!CLK * D)	0.02299	0.04733	0.41262
	hold	SN (R)	(!CLK * !D)	0.02052	0.04733	0.41944
	setup	SN (R)	(!CLK * D)	0.02195	0.03389	1.19135
	setup	SN (R)	(!CLK * !D)	0.01873	0.02989	0.94685

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	RN ()	(!CLK * D * SN)	0.06335	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06335	0.40527	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQNX1	hold	RN (R)	0.03456	0.06477	0.33886
	setup	RN (R)	-0.01427	-0.02242	0.35650

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	hold	RN (R)	(!CLK * D)	0.03456	0.06477	0.32267
	hold	RN (R)	(!CLK * !D)	0.03456	0.06477	0.33886
	setup	RN (R)	(!CLK * D)	-0.01427	-0.02242	0.35650
	setup	RN (R)	(!CLK * !D)	-0.01580	-0.02242	0.04339

Constraints(ns) for SN falling (conditional):

[illegible]

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.13728	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.19581	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	65759.40000	65742.50000	65564.10000
	RN	62319.10000	62155.50000	61909.00000

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	32.48890	32.48880	32.69810
	RN	32.73270	32.73280	33.13210
	SN	-0.00000	-0.00000	-0.00000
	SN	33.55220	33.58250	34.23850

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(RN * !SN * !QN)	0.00000	0.00000	0.00000
	(RN * !SN * !QN)	32.64280	32.64250	32.84360
	(CLK * RN * SN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * QN)	154979.00000	154956.00000	154953.00000
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	59694.30000	59694.10000	59694.20000
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	19.47050	19.50460	20.28800
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	61118.40000	61119.10000	61118.40000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(RN * !SN * !QN)	0.00000	0.00000	0.00000
	(RN * !SN * !QN)	15.80820	15.81030	16.03130
	(CLK * RN * SN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * QN)	147665.00000	147637.00000	147597.00000
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	59242.30000	59389.60000	59426.50000
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	50209.20000	50209.00000	50205.80000
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	61263.00000	61274.20000	61279.60000

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * QN)	136977.00000	136973.00000	136976.00000
	(CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * QN)	130337.00000	130297.00000	130311.00000
	(!CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * QN)	19.29720	19.32970	20.09150
	(!CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * QN)	62347.10000	62347.60000	62347.80000

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * QN)	55979.20000	55976.50000	55981.60000
	(CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * QN)	55517.60000	55517.80000	55521.80000
	(!CLK * D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * QN)	49145.10000	49160.80000	49156.30000
	(!CLK * !D * SN * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * QN)	60073.70000	60075.30000	60078.90000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * !QN)	30.74060	30.74120	30.74250
	(!RN * QN)	0.00000	0.00000	0.00000
	(!RN * QN)	55283.00000	55276.40000	55279.30000
	(!CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * !QN)	16.86570	16.86530	16.86660
	(!CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * !QN)	54348.90000	54353.60000	54351.70000

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * !QN)	31.99880	32.00320	32.00470
	(!RN * QN)	0.00000	0.00000	0.00000
	(!RN * QN)	7.00236	7.03567	7.87801
	(!CLK * D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * !QN)	18.06590	18.06690	18.06800
	(!CLK * !D * RN * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * !QN)	7.95489	7.98287	8.60381

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(D * RN * SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !QN)	32.44890	32.44790	32.64810
	(D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * !QN)	32.44550	32.44470	32.64240
	(!RN * SN * QN)	0.00000	0.00000	0.00000
	(!RN * SN * QN)	57886.50000	57893.10000	57945.20000
	(!RN * !SN * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * QN)	3.96868	3.96752	4.11502
	(!D * RN * SN * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * QN)	148453.00000	148419.00000	148406.00000
	(!D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * !QN)	17.87620	17.87840	18.13660

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQNX1	(D * RN * SN * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * QN)	18.98860	19.02810	19.82590
	(D * RN * SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !QN)	16.19800	16.20470	16.43490
	(D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * !QN)	16.19020	16.19860	16.41770
	(!RN * SN * QN)	0.00000	0.00000	0.00000
	(!RN * SN * QN)	61269.50000	61269.80000	61269.20000
	(!RN * !SN * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * QN)	0.08043	0.08386	0.28934
	(!D * RN * SN * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * QN)	61875.10000	61876.50000	61876.70000
	(!D * RN * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * !QN)	5.85766	5.86368	6.14656

DFFSNRNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0
x	1	1	x	IQ

Footprint

Cell Name	Area
DFFSNRNQX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
DFFSNRNQX1	0.01012	0.03402	0.02266	0.02233	3.55003

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RR)	0.15525	0.25226	0.42628
	SN->Q (FR)	0.03607	0.14614	0.80584

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RF)	0.22635	0.31329	0.41092
	RN->Q (FF)	0.13156	0.27850	1.04070
	SN->Q (RF)	0.05213	0.10798	0.03128

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.01272	-0.00923	2.06419
	setup	CLK (R)	0.16941	0.20032	0.71621

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.06166	-0.11211	-1.08492
	setup	CLK (R)	0.10537	0.15944	2.33314

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.01272	-0.00923	2.06419
	setup	CLK (R)	(RN * SN)	0.16941	0.20032	0.71621

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06166	-0.11211	-1.08492
	setup	CLK (R)	(RN * SN)	0.10537	0.15944	2.33314

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.15976	0.20733	7.78128
	removal	CLK (R)	-0.01430	-0.01993	-0.06334
	hold	SN (R)	0.01087	0.02491	0.15394
	setup	SN (R)	0.01873	0.02989	1.00625

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.15976	0.20733	7.78128
	removal	CLK (R)	(D * SN)	-0.01430	-0.01993	-0.06334
	hold	SN (R)	(CLK * D)	0.00534	-0.00996	-0.10209
	hold	SN (R)	(CLK * !D)	0.00381	-0.00996	-0.10466
	hold	SN (R)	(!CLK * D)	0.01087	0.02491	0.14626
	hold	SN (R)	(!CLK * !D)	0.01087	0.02491	0.15394
	setup	SN (R)	(CLK * D)	0.00922	0.01993	0.56914
	setup	SN (R)	(CLK * !D)	0.00823	0.01993	0.66473
	setup	SN (R)	(!CLK * D)	0.01873	0.02989	1.00625
	setup	SN (R)	(!CLK * !D)	0.01873	0.02989	0.82958

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	RN ()	(CLK * D * SN)	0.10956	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.09724	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.06335	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06335	0.40527	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.02542	0.02005	4.24543
	removal	CLK (R)	0.00273	0.00498	0.01140
	hold	RN (R)	0.04756	0.09467	0.65404
	setup	RN (R)	0.00859	-0.00400	0.64047

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(!D * RN)	0.02542	0.02005	4.24543
	removal	CLK (R)	(!D * RN)	0.00273	0.00498	0.01140
	hold	RN (R)	CLK	0.04413	0.09467	0.65404
	hold	RN (R)	(!CLK * D)	0.04756	0.09218	0.59482
	hold	RN (R)	(!CLK * !D)	0.04756	0.09218	0.61284
	setup	RN (R)	CLK	0.00859	-0.00400	0.64047
	setup	RN (R)	(!CLK * D)	-0.01580	-0.02242	0.41017
	setup	RN (R)	(!CLK * !D)	-0.01331	-0.02491	0.06202

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08492	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08492	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.06951	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.06951	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.15268	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23277	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11572	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.67160	32.67070	32.87950
	SN	32.59770	32.60590	32.94870

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.50640	32.50550	32.72140
	RN	-0.00000	-0.00000	-0.00000
	RN	33.06250	33.06700	33.43230
	SN	32.67620	32.67810	33.07530

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	32.63850	32.63810	32.83720
	$(CLK * RN * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q)$	31.10860	31.11060	31.11060
	$(CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q)$	32.63710	32.63680	32.83750
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	31.09980	31.10040	31.10100
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	17.56850	17.56940	17.76560
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	17.19780	17.19940	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	15.79960	15.80150	16.01610
	$(CLK * RN * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q)$	31.73540	31.73460	31.73490
	$(CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q)$	15.80170	15.80410	16.02450
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	31.72860	31.72790	31.72880
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	17.58360	17.58760	17.80420
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	17.79790	17.79820	17.79850

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	30.36420	30.36390	30.36250
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	32.88970	32.88960	33.27330
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	30.36090	30.35900	30.35860
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	17.62570	17.62320	17.90980
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	16.91790	16.91510	17.10820
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	14.12340	14.12010	14.46640
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	16.50770	16.50760	16.50620
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	3.61102	3.61066	3.78112

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	32.24140	32.24340	32.24330
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	4.84158	4.85229	5.46297
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	32.24160	32.24410	32.24480
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	6.13316	6.13710	6.54180
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	18.12570	18.12770	18.32370
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	0.10950	0.12056	0.55824
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	18.30460	18.30300	18.30280
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	0.08313	0.08827	0.28323

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	30.73420	30.73760	30.74070
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	16.85930	16.86330	16.86550
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	16.87160	16.86800	16.87170
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	17.58740	17.58760	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	31.98860	31.99620	31.99740
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	18.05970	18.06440	18.06460
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	18.06270	18.06450	18.06470
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	6.07005	6.07428	6.28628

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	32.63510	32.63290	32.82960
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	32.63370	32.63300	32.82630
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	32.44790	32.44700	32.64070
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	3.97148	3.96945	4.11832
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	32.45710	32.45620	32.65370
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	18.13620	18.13830	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	15.81260	15.82000	16.06380
	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	16.61300	16.61420	16.82590
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	15.80860	15.81440	16.03970
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	16.18980	16.18920	16.39260
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	0.08369	0.08717	0.29352
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	17.60100	17.60300	17.85610
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	16.19240	16.19110	16.40700
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	5.63984	5.64527	5.93703

DFFSNRNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT	
D	RN	SN	CLK	Q	QN
0	1	1	R	0	1
1	1	1	R	1	0
x	0	0	x	1	1
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNRNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	D	RN	SN	CLK	Q	QN
DFFSNRNX1	0.01012	0.03429	0.02266	0.02233	3.55003	3.57679

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RR)	0.15528	0.25228	0.42624
	QN->Q (FR)	0.04460	0.15773	0.86773
	SN->Q (FR)	0.03607	0.14614	0.80584

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RF)	0.22628	0.31321	0.41082
	QN->Q (RF)	0.06600	0.09856	-0.21304
	RN->Q (FF)	0.13152	0.27847	1.04058
	SN->Q (RF)	0.05214	0.10798	0.03128

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RR)	0.15086	0.23771	0.33094
	Q->QN (FR)	0.03297	0.12084	0.39045
	RN->QN (FR)	0.03500	0.14155	0.83112

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RF)	0.22064	0.31885	0.51702
	Q->QN (RF)	0.04801	0.11392	-0.00012
	RN->QN (RF)	0.24800	0.36883	0.76343
	SN->QN (FF)	0.31594	0.42626	1.06611

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.01271	-0.00924	2.06422
	setup	CLK (R)	0.16940	0.20039	0.71893

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.06175	-0.11211	-1.08500
	setup	CLK (R)	0.10540	0.15944	2.34219

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.01271	-0.00924	2.06422
	setup	CLK (R)	(RN * SN)	0.16940	0.20039	0.71893

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.06175	-0.11211	-1.08500
	setup	CLK (R)	(RN * SN)	0.10540	0.15944	2.34219

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.15966	0.20765	7.76843
	removal	CLK (R)	-0.01430	-0.01993	-0.06334
	hold	SN (R)	0.02146	0.04982	0.41088
	setup	SN (R)	0.01873	0.02989	1.00520

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(D * SN)	0.15966	0.20765	7.76843
	removal	CLK (R)	(D * SN)	-0.01430	-0.01993	-0.06334
	hold	SN (R)	(CLK * D)	0.00534	-0.00996	-0.10209
	hold	SN (R)	(CLK * !D)	0.00381	-0.00996	-0.10466
	hold	SN (R)	(!CLK * D)	0.02143	0.04982	0.40530
	hold	SN (R)	(!CLK * !D)	0.02146	0.04733	0.41088
	setup	SN (R)	(CLK * D)	0.00922	0.01993	0.57086
	setup	SN (R)	(CLK * !D)	0.00813	0.01993	0.67779
	setup	SN (R)	(!CLK * D)	0.01873	0.02989	1.00520
	setup	SN (R)	(!CLK * !D)	0.01873	0.02989	0.82853

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.10956	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.09724	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.06335	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.06335	0.40527	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.02534	0.02013	4.19396
	removal	CLK (R)	-0.00127	0.00498	0.01140
	hold	RN (R)	0.04756	0.09467	0.65404
	setup	RN (R)	0.00862	-0.00400	0.64063

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(!D * RN)	0.02534	0.02013	4.19396
	removal	CLK (R)	(!D * RN)	-0.00127	0.00498	0.01140
	hold	RN (R)	CLK	0.04413	0.09467	0.65404
	hold	RN (R)	(!CLK * D)	0.04509	0.09218	0.59482
	hold	RN (R)	(!CLK * !D)	0.04756	0.09218	0.61284
	setup	RN (R)	CLK	0.00862	-0.00400	0.64063
	setup	RN (R)	(!CLK * D)	-0.01580	-0.02242	0.41025
	setup	RN (R)	(!CLK * !D)	-0.01331	-0.02491	0.06209

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08492	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08492	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.06951	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.06951	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRN1	min_pulse_width	CLK ()	(D * RN * SN)	0.15268	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14036	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRN1	min_pulse_width	CLK ()	(D * RN * SN)	0.23277	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11572	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33580	16.33520	16.43980
	SN	-0.00000	-0.00000	-0.00000
	SN	16.29910	16.30270	16.47430

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25320	16.25250	16.36070
	RN	-0.00000	-0.00000	-0.00000
	RN	16.53120	16.53350	16.71610
	SN	32.67620	32.67810	33.07530

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25290	16.25280	16.36090
	RN	-0.00000	-0.00000	-0.00000
	RN	16.53070	16.53340	16.71320

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33620	16.33520	16.43970
	RN	32.89160	32.89140	33.27970
	SN	-0.00000	-0.00000	-0.00000
	SN	16.29940	16.30320	16.47600

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	32.63850	32.63810	32.83720
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.10860	31.11060	31.11060
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	32.63750	32.63710	32.83790
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.09980	31.10040	31.10090
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.57320	17.57320	17.77380
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.19760	17.19930	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	15.79960	15.80150	16.01610
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.73540	31.73450	31.73490
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	15.80170	15.80410	16.02450
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.72880	31.72820	31.72900
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.58840	17.59180	17.81410
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.79780	17.79820	17.79840

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * D * SN * !Q * QN)$	30.36400	30.36390	30.36250
	$(CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !D * SN * !Q * QN)$	30.36090	30.35900	30.35860
	$(!CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q * QN)$	16.91790	16.91520	17.10820
	$(!CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !D * SN * !Q * QN)$	16.50760	16.50760	16.50610

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	32.24130	32.24340	32.24320
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	32.24160	32.24410	32.24480
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	18.12570	18.12770	18.32360
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	18.30450	18.30300	18.30280

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	30.73420	30.73760	30.74070
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	16.86160	16.86330	16.86550
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	16.87160	16.86800	16.87170
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	17.58740	17.58760	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	31.98860	31.99620	31.99740
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	18.05970	18.06440	18.06460
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	18.06270	18.06450	18.06470
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	6.06983	6.07409	6.28575

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	32.63510	32.63300	32.82970
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	32.63380	32.63310	32.82630
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	32.44790	32.44700	32.64070
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	3.97147	3.96943	4.11830
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	32.45710	32.45610	32.65360
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	18.13620	18.13830	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	15.81260	15.82000	16.06380
	(D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q * QN)	16.61270	16.61380	16.82550
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	15.80860	15.81410	16.03970
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	16.18980	16.18920	16.39260
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	0.08367	0.08715	0.29350
	(!D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q * !QN)	17.60090	17.60300	17.85600
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	16.19230	16.19080	16.40660
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	5.63982	5.64525	5.93703

DFFSNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT	
D	SN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	1	0
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	SN	CLK	Q	QN
DFFSNX1	0.01015	0.02167	0.02229	3.67547	5.23618

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RR)	0.10878	0.21340	0.25654
	QN->Q (FR)	0.04398	0.15780	0.90278
	SN->Q (FR)	0.03552	0.14540	0.80783

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RF)	0.21264	0.29928	0.40797
	QN->Q (RF)	0.06509	0.09924	-0.20670

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RR)	0.13836	0.22463	0.32673
	Q->QN (FR)	0.02979	0.11600	0.52142

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RF)	0.15499	0.26097	0.32403
	Q->QN (RF)	0.03096	0.07771	-0.28239
	SN->QN (FF)	0.24126	0.35811	1.00372

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	0.01232	0.01593	0.82179
	setup	CLK (R)	0.13195	0.17388	4.30899

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	-0.06793	-0.11958	-1.15292
	setup	CLK (R)	0.10206	0.15744	1.62351

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	0.01232	0.01593	0.82179
	setup	CLK (R)	SN	0.13195	0.17388	4.30899

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	-0.06793	-0.11958	-1.15292
	setup	CLK (R)	SN	0.10206	0.15744	1.62351

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	recovery	CLK (R)	0.03321	0.03586	4.44830
	removal	CLK (R)	-0.01681	-0.01495	-0.12337

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	recovery	CLK (R)	!D	0.03321	0.03586	4.44830
	removal	CLK (R)	!D	-0.01681	-0.01495	-0.12337

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	SN ()	(CLK * D)	0.06643	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.06643	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.06027	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06027	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.10032	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13112	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.20505	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10648	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.82029	9.82016	9.93725
	SN	-0.00000	-0.00000	-0.00000
	SN	9.49982	9.50361	9.67955

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68140	17.68110	17.79400

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68140	17.68120	17.79390

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.82015	9.82014	9.93796
	SN	-0.00000	-0.00000	-0.00000
	SN	9.49995	9.50332	9.68462

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	19.80290	19.80230	20.00770
	$(CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q * QN)$	33.71140	33.71230	33.71360
	$(CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q * !QN)$	19.79770	19.79890	20.00590
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	10.28740	10.28890	10.49310

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	9.55577	9.55809	9.77689
	$(CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q * QN)$	34.35860	34.35990	34.36010
	$(CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q * !QN)$	9.55975	9.56251	9.78781
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	14.91860	14.92420	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	18.32530	18.32660	18.33250
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	9.86823	9.87388	9.87330
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	9.87741	9.87449	9.87827
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	14.41070	14.41020	14.60650

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	19.54940	19.55510	19.55620
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	11.02420	11.02730	11.02770
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	11.02810	11.02710	11.02780
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	3.67812	3.68265	3.89182

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	19.79670	19.79620	20.01960
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	19.79250	19.79200	20.01260
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	35.14040	35.13990	35.34260
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	10.63760	10.64350	10.91050

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	9.56713	9.57466	9.85099
	(D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * SN * !Q * QN)	13.00550	13.00390	13.24420
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	9.55751	9.56655	9.81662
	(!D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * SN * Q * !QN)	14.92510	14.92720	15.17990
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	17.39310	17.39310	17.60770
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	3.48942	3.49446	3.80088

DFFX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	R	0	1
1	R	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
DFFX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
DFFX1	0.01055	0.02318	5.42033	5.42584

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RR)	0.11293	0.22449	0.33742
	QN->Q (FR)	0.03633	0.13727	0.79002

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RF)	0.19112	0.28010	0.40053
	QN->Q (RF)	0.03998	0.07666	-0.27623

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RR)	0.14045	0.22896	0.34018
	Q->QN (FR)	0.03016	0.11456	0.52082

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RF)	0.15926	0.27211	0.40534
	Q->QN (RF)	0.03192	0.08157	-0.21128

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	0.04560	0.07083	0.74072
	setup	CLK (R)	0.10806	0.15058	0.81709

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	-0.03337	-0.06975	-0.73902
	setup	CLK (R)	0.05923	0.09965	1.54647

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.10340	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.13112	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.19273	0.40527	16.50020
	min_pulse_width	CLK ()	!D	0.09108	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56560	10.56550	10.68070

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12850	13.12790	13.23860

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12800	13.12800	13.23870

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56520	10.56570	10.68090

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	$(CLK * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * Q * !QN)$	21.09480	21.09540	21.31620
	$(CLK * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !Q * QN)$	24.93700	24.93590	24.93790
	$(!CLK * Q * !QN) + (!CLK * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * Q * !QN) + (!CLK * !Q * QN)$	11.29670	11.29880	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	$(CLK * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * Q * !QN)$	10.10460	10.10790	10.34910
	$(CLK * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !Q * QN)$	25.56830	25.56880	25.56920
	$(!CLK * Q * !QN) + (!CLK * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * Q * !QN) + (!CLK * !Q * QN)$	11.30910	11.31320	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	$(D * Q * !QN)$	0.00000	0.00000	0.00000
	$(D * Q * !QN)$	21.09780	21.09630	21.30850
	$(!D * !Q * QN)$	0.00000	0.00000	0.00000
	$(!D * !Q * QN)$	26.21620	26.21560	26.41820

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	10.11690	10.11990	10.35220
	(D * !Q * QN)	0.00000	0.00000	0.00000
	(D * !Q * QN)	10.32750	10.32800	10.54020
	(!D * Q * !QN)	0.00000	0.00000	0.00000
	(!D * Q * !QN)	11.32050	11.32520	11.55020
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	9.90474	9.90814	10.11400

DLATCHN

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Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE_N	Q
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
DLATCHN	179.87680

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE_N	Q
DLATCHN	0.02246	0.01036	2.88873

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCHN	0.00000	22.21240	28.71060

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (RR)	0.16052	0.22929	0.11715
	GATE_N->Q (-R)	0.21067	0.33489	1.07308

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (FF)	0.14754	0.25030	0.91467
	GATE_N->Q (-F)	0.16260	0.29140	1.04068

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	13.54480	13.55390	14.05040
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	13.29380	13.29820	13.54750

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	9.19627	9.21044	9.69568
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	7.79058	7.79694	8.04686

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	16.22130	16.22520	16.47220

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	11.66590	11.67170	11.91190

Passive power(pJ) for GATE_N rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	16.31900	16.32340	16.59530
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	11.95920	11.96010	12.20380

Passive power(pJ) for GATE_N falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	11.77350	11.77970	12.03900
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.99034	9.00012	9.25036

DLATCH

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Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE	Q
x	0	1
0	1	0
1	1	1

Footprint

Cell Name	Area
DLATCH	162.64960

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE	Q
DLATCH	0.02245	0.02034	2.90667

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCH	0.00000	18.14490	24.27320

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (RR)	0.16009	0.22929	0.13291
	GATE->Q (-R)	0.16016	0.22079	0.01872

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (FF)	0.14606	0.24839	0.90585
	GATE->Q (-F)	0.10679	0.18216	0.14120

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	14.12830	14.13890	14.64010
	GATE	0.00000	0.00000	0.00000
	GATE	13.88760	13.88930	14.12120

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	9.55235	9.56902	10.05510
	GATE	0.00000	0.00000	0.00000
	GATE	8.12131	8.12575	8.34771

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	12.15600	12.16090	12.40740

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	8.01476	8.02208	8.26039

Passive power(pJ) for GATE rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.37440	12.37790	12.61050
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.32346	9.32823	9.55308

Passive power(pJ) for GATE falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.18790	12.19510	12.45500
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.23606	8.24347	8.47690

FA

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT	
A	B	CIN	COUT	SUM
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
FA	309.08081

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	A	B	CIN	COUT	SUM
FA	0.03793	0.03734	0.03457	5.74192	2.61746

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
FA	0.00000	36.98230	51.95610

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (RR)	0.30815	0.36076	0.09770
	B->COUT (RR)	0.33215	0.38714	0.15370
	CIN->COUT (RR)	0.13166	0.19261	0.04427

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (FF)	0.27995	0.38567	1.00972
	B->COUT (FF)	0.28968	0.39534	1.02277
	CIN->COUT (FF)	0.15229	0.27009	0.99735

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-R)	-	0.23250	0.36714	1.54223
	B->SUM (-R)	-	0.21486	0.34499	1.33814
	CIN->SUM (RR)	$(A * B) + (!A * !B)$	0.08359	0.14025	-0.05048
	CIN->SUM (FR)	$(A * !B) + (!A * B)$	0.04533	0.11877	0.78235

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-F)	-	0.20785	0.34431	1.57620
	B->SUM (-F)	-	0.19032	0.32262	1.37133
	CIN->SUM (FF)	$(A * B) + (!A * !B)$	0.06770	0.16748	0.72135
	CIN->SUM (RF)	$(A * !B) + (!A * B)$	0.02782	0.08350	-0.13508

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	17.40010	17.40640	17.77430
	B	17.39940	17.40560	17.77550
	CIN	0.00000	0.00000	0.00000
	CIN	14.35110	14.35780	14.72720

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	5.75267	5.75419	5.88584
	B	5.75538	5.75745	5.88073
	CIN	0.00000	0.00000	0.00000
	CIN	3.75755	3.76547	4.09840

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	26.85690	26.86980	27.53710
	B	-	0.00000	0.00000	0.00000
	B	-	26.85430	26.86900	27.54130
	CIN	$(A * B) + (!A * !B)$	0.00000	0.00000	0.00000
	CIN	$(A * B) + (!A * !B)$	27.03870	27.04420	27.28200
	CIN	$(A * !B) + (!A * B)$	0.00000	0.00000	0.00000
	CIN	$(A * !B) + (!A * B)$	3.75696	3.76519	4.09762

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	28.06380	28.08240	28.78190
	B	-	0.00000	0.00000	0.00000
	B	-	28.06220	28.08010	28.76090
	CIN	$(A * B) + (!A * !B)$	0.00000	0.00000	0.00000
	CIN	$(A * B) + (!A * !B)$	33.67450	33.68110	33.91010
	CIN	$(A * !B) + (!A * B)$	0.00000	0.00000	0.00000
	CIN	$(A * !B) + (!A * B)$	14.35040	14.35760	14.72670

HA

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp
25.00*

Truth Table

INPUT		OUTPUT	
A	B	COUT	SUM
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Footprint

Cell Name	Area
HA	136.80881

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	A	B	COUT	SUM
HA	0.03355	0.03772	6.17943	2.60808

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
HA	0.00000	17.03700	32.46250

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (RR)	0.05352	0.10823	0.17097
	B->COUT (RR)	0.04779	0.11303	0.28511

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (FF)	0.04842	0.15145	0.90285
	B->COUT (FF)	0.04497	0.13385	0.54959

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (RR)	!B	0.05780	0.11244	-0.17922
	A->SUM (FR)	B	0.05149	0.14062	1.20954
	B->SUM (RR)	!A	0.08368	0.12975	-0.21619
	B->SUM (FR)	A	0.04352	0.11177	0.70210

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (FF)	!B	0.06011	0.16041	0.78473
	A->SUM (RF)	B	0.03008	0.07320	-0.10167
	B->SUM (FF)	!A	0.06813	0.16968	0.75532
	B->SUM (RF)	A	0.02566	0.07669	-0.00675

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	11.85330	11.86360	12.34290
	B	0.00000	0.00000	0.00000
	B	11.85090	11.86420	12.36020

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	0.86476	0.87728	1.30813
	B	0.00000	0.00000	0.00000
	B	0.85628	0.86631	1.20062

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	0.86491	0.87674	1.30758
	A	!B	0.00000	0.00000	0.00000
	A	!B	3.99664	4.00053	4.24177
	B	A	0.00000	0.00000	0.00000
	B	A	0.85637	0.86619	1.20048
	B	!A	0.00000	0.00000	0.00000
	B	!A	4.01579	4.01460	4.26669

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	11.85100	11.86580	12.34160
	A	!B	0.00000	0.00000	0.00000
	A	!B	9.43971	9.44678	9.68547
	B	A	0.00000	0.00000	0.00000
	B	A	11.85120	11.86400	12.35850
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.44585	9.45235	9.65950

INVX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
INVX1	24.83200

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
INVX1	0.01040	5.57495

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
INVX1	0.00000	3.90439	7.80687

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (FR)	0.01250	0.05151	0.52048

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (RF)	0.00968	-0.01123	-0.71148

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	0.01791	0.03282	0.36197

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	4.60745	4.62088	4.94936

INV

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp
25.00*

Truth Table

INPUT	OUTPUT
A	Y
x	0

Footprint

Cell Name	Area
inv	0.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
inv	0.00052	1.80000

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
inv	0.00000	0.00000	0.00000

MUX2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT	OUTPUT
S	Y
x	1

Footprint

Cell Name	Area
MUX2X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	S	Y
MUX2X1	0.02099	6.62826

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
MUX2X1	0.00000	13.86590	17.35870

NAND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
NAND2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NAND2X1	0.01054	0.01061	5.10239

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND2X1	0.00000	3.89237	15.56580

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (FR)	0.01995	0.09545	1.02849
	B->Y (FR)	0.01558	0.05743	0.31595

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (RF)	0.02205	0.02847	-0.35838
	B->Y (RF)	0.01600	0.02474	-0.35906

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	0.02839	0.04125	0.39106
	B	0.00000	0.00000	0.00000
	B	0.01844	0.02819	0.26197

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	9.73378	9.74025	9.98852
	B	0.00000	0.00000	0.00000
	B	9.73388	9.74232	10.01070

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	-0.00179	-0.00183	-0.00180

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01113	0.01061	0.01036

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.00105	0.00100	0.00101

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.01104	0.01124	0.01127

NAND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
NAND3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NAND3X1	0.01059	0.01035	0.01056	3.37535

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND3X1	0.00000	2.91632	23.32440

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (FR)	0.02578	0.11740	1.16502
	B->Y (FR)	0.02327	0.09439	0.60842
	C->Y (FR)	0.01803	0.06390	0.18449

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (RF)	0.03862	0.06030	-0.16612
	B->Y (RF)	0.03257	0.05689	-0.16633
	C->Y (RF)	0.02446	0.05498	-0.10550

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	0.03811	0.04922	0.39511
	B	0.00000	0.00000	0.00000
	B	0.02917	0.03734	0.28647
	C	0.00000	0.00000	0.00000
	C	0.01993	0.02699	0.22526

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	14.72020	14.72240	14.94240
	B	0.00000	0.00000	0.00000
	B	14.72100	14.72350	14.94930
	C	0.00000	0.00000	0.00000
	C	14.72010	14.72500	14.95950

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	-0.00240	-0.00243	-0.00243
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.00068	0.00064	0.00062
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	-0.00450	-0.00454	-0.00456

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.01152	0.01078	0.01061
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.01292	0.01241	0.01225
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	0.00998	0.00993	0.00995

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	-0.00211	-0.00213	-0.00219
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.00425	0.00421	0.00423
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	-0.00468	-0.00471	-0.00473

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.01112	0.01044	0.01029
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.01201	0.01213	0.01216
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	0.01101	0.01114	0.01116

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.00018	0.00012	0.00010
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.00448	0.00442	0.00444

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01100	0.01119	0.01122
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.01079	0.01093	0.01089

NOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
x	1	0
1	x	0

Footprint

Cell Name	Area
NOR2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NOR2X1	0.01081	0.01051	2.54971

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR2X1	0.00000	2.00110	6.04921

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (FR)	0.03417	0.09468	0.84143
	B->Y (FR)	0.02202	0.08126	0.78326

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (RF)	0.01428	0.03255	-0.37506
	B->Y (RF)	0.01191	-0.00019	-0.79535

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	0.03254	0.03624	0.21318
	B	0.00000	0.00000	0.00000
	B	0.02693	0.03614	0.27672

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	2.45459	2.46347	2.74835
	B	0.00000	0.00000	0.00000
	B	2.72005	2.72571	2.92822

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.06865	0.06688	0.06652

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	3.67718	3.67728	3.67695

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	0.01664	0.01663	0.01663

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	2.45798	2.45736	2.45805

NOR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT
A
x

Footprint

Cell Name	Area
NOR3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)
	A
NOR3X1	51.68590

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR3X1	0.00000	24400400.00000	48800800.00000

Passive Power Information

Passive power(pJ) for A rising :

Cell Name	Power(pJ)		
	first	mid	last
NOR3X1	0.00000	0.00000	0.00000
	20.89850	25.30130	106.88300

Passive power(pJ) for A falling :

Cell Name	Power(pJ)		
	first	mid	last
NOR3X1	0.00000	0.00000	0.00000
	24185200.00000	24185200.00000	24184900.00000

OR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
OR2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
OR2X1	0.01056	0.01043	5.70062

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR2X1	0.00000	3.08382	5.40294

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (RR)	0.04053	0.10366	-0.02116
	B->Y (RR)	0.03668	0.08105	-0.38150

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (FF)	0.07668	0.17881	1.21596
	B->Y (FF)	0.06329	0.17170	1.21027

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.85473	1.86425	2.15363
	B	0.00000	0.00000	0.00000
	B	2.93805	2.94271	3.18544

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.25467	1.25606	1.44939
	B	0.00000	0.00000	0.00000
	B	2.95795	2.96640	3.22200

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.04935	0.04753	0.04726

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	3.12333	3.12275	3.12350

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	0.01973	0.01970	0.01974

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	1.84556	1.84538	1.84610

OR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	0
0	x	1	1
x	1	x	1
1	x	x	1

Footprint

Cell Name	Area
OR3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
OR3X1	0.01070	0.01038	0.01038	5.59917

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR3X1	0.00000	1.61262	5.01932

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (RR)	0.04361	0.11690	0.01045
	B->Y (RR)	0.04318	0.10599	-0.27102
	C->Y (RR)	0.03851	0.08463	-0.50843

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (FF)	0.12454	0.22039	1.35914
	B->Y (FF)	0.11279	0.22065	1.43401
	C->Y (FF)	0.09002	0.20905	1.42914

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	1.89598	1.90442	2.20505
	B	0.00000	0.00000	0.00000
	B	1.61696	1.62225	1.83353
	C	0.00000	0.00000	0.00000
	C	2.55330	2.55798	2.75771

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	0.62728	0.62730	0.79118
	B	0.00000	0.00000	0.00000
	B	1.31322	1.31520	1.48328
	C	0.00000	0.00000	0.00000
	C	2.58977	2.59622	2.81556

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01839	0.01750	0.01735
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.06621	0.06453	0.06428
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.04955	0.04768	0.04731

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06097	0.06108	0.06113
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1.53493	1.53524	1.53528
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	2.67781	2.67764	2.67786

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02081	0.02081	0.02087
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.03900	0.03900	0.03900
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.03935	0.03725	0.03698

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06466	0.06467	0.06470
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	1.87992	1.87978	1.88012
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	2.75176	2.75130	2.75146

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.01258	0.01256	0.01256
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	0.01730	0.01728	0.01728

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.08814	0.08816	0.08819
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	1.88340	1.88332	1.88369

TIEHI

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIEHI	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
TIEHI	12.24737

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIEHI	0.00000	0.00000	0.00000

TIELO

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIELO	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	YN
TIELO	26.69489

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIELO	0.00000	0.00000	0.00000

TMRDFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80, Temp
25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
x	x	IQN
1	1	IQN

Footprint

Cell Name	Area
TMRDFFQNX1	584.71600

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
TMRDFFQNX1	0.04010	0.07455	2.70083

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQNX1	0.00000	71.39310	102.09600

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RR)	0.36008	0.45131	0.68631

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	0.04249	0.06687	0.59795
	setup	CLK (R)	-0.03465	-0.05232	-0.22592

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	-0.02294	-0.04482	-0.49083
	setup	CLK (R)	0.05321	0.08591	1.03286

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	!D	0.14344	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	!D	0.09108	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	76.32800	76.37050	77.68560

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	72660.80000	72661.90000	72661.90000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	72248.70000	72249.90000	72261.00000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71980.50000	71983.10000	71985.30000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	64003.20000	64003.80000	64004.40000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	64579.80000	64579.50000	64582.40000
	!CLK	0.00000	0.00000	0.00000
	!CLK	89971.40000	89974.40000	89967.90000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	63371.40000	63384.30000	63406.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	65506.90000	66064.30000	66661.70000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	76.35760	76.39620	77.69390

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	103219.00000	103817.00000	100654.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	98414.30000	98484.30000	98720.60000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	98893.80000	98877.80000	98878.40000
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	88901.50000	88899.20000	89077.10000

TMRDFFQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	Q
x	x	-

Footprint

Cell Name	Area
TMRDFFQX1	601.94318

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
TMRDFFQX1	0.04023	0.07448	8.45426

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQX1	0.00000	77.76220	108.74900

Delay Information

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	CLK->Q (RF)	0.42652	0.51779	0.74874

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	0.04459	0.06379	0.56785
	setup	CLK (R)	-0.03465	-0.05232	-0.22592

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	-0.02294	-0.04482	-0.49831
	setup	CLK (R)	0.04931	0.08640	1.02036

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	!D	0.14344	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	!D	0.08800	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	-0.00000	-0.00000	-0.00000
	CLK	80.66380	80.70610	82.02320

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	72259.80000	72260.60000	72270.00000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	72295.00000	72294.40000	72293.90000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71607.60000	71606.20000	71608.80000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	64324.60000	64323.40000	64328.70000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	63863.40000	63863.00000	63864.20000
	!CLK	0.00000	0.00000	0.00000
	!CLK	87405.00000	87397.60000	87408.80000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	65802.80000	66334.80000	66895.90000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	63399.90000	63414.40000	63431.90000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	80.63340	80.67230	81.96450

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	98605.50000	98677.30000	98936.50000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	103414.00000	103955.00000	100553.00000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	88935.00000	88932.10000	89130.30000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	99586.30000	99573.10000	99580.60000

TMRDFFRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80,
Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
x	x	x	-

Footprint

Cell Name	Area
TMRDFFRNQX1	705.30640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
TMRDFFRNQX1	0.04040	0.09156	0.06856	0.50314

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFRNQX1	0.00000	763123.00000	1056050.00000

Delay Information

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFRNQX1	CLK->Q (FF)	524.02300	524.12700	523.94200
	D->Q (RF)	0.70334	0.89039	3.76295

Constraints(ns) for CLK falling (conditional):

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Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	CLK	-0.00000	-0.00000	-0.00000
	CLK	531300.00000	531300.00000	531313.00000
	D	-0.00000	-0.00000	-0.00000
	D	473434.00000	473432.00000	473433.00000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	427441.00000	427440.00000	427437.00000
	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	473827.00000	473826.00000	473827.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	531748.00000	531748.00000	531751.00000

TMRDFFSNQNX1

sky130_rhbd_tt_1P8_25C.ccs
Cell Library: Process , Voltage
1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
x	0	x	1
-	1	-	HiZ

Footprint

Cell Name	Area
TMRDFFSNQNX1	653.62482

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
TMRDFFSNQNX1	0.04124	0.06902	0.07193	2.60418

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQNX1	0.00000	83.86200	158.19400

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQNX1	CLK->QN (RR)	0.31232	0.42152	0.51786
	SN->QN (FR)	0.38418	0.50116	1.11729

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQNX1	CLK->QN (RF)	0.21711	0.30729	0.48248

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	0.01278	0.01593	0.70846
	setup	CLK (R)	0.11689	0.13771	0.96567

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	-0.07166	-0.12456	-1.11139
	setup	CLK (R)	0.10095	0.15944	1.64264

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	SN	0.01278	0.01593	0.70846
	setup	CLK (R)	SN	0.11689	0.13771	0.96567

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	hold	CLK (R)	SN	-0.07166	-0.12456	-1.11139
	setup	CLK (R)	SN	0.10095	0.15944	1.64264

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQNX1	recovery	CLK (R)	0.03418	0.03564	0.29907
	removal	CLK (R)	-0.01435	-0.01495	-0.11640

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	recovery	CLK (R)	!D	0.03418	0.03564	0.29907
	removal	CLK (R)	!D	-0.01435	-0.01495	-0.11640

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	SN ()	(CLK * D)	0.07259	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07259	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.08800	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.08800	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.11880	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.17733	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.18965	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10956	0.40527	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	56.58070	56.58470	57.28320
	SN	54.97800	54.99680	56.03010

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	105.16900	105.16500	105.78100

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(CLK * SN) + (CLK * !SN * QN) + (!CLK * !SN * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * QN) + (!CLK * !SN * QN)$	100.67500	100.67600	100.67600
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	32.81800	32.82350	33.42850

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(CLK * SN) + (CLK * !SN * QN) + (!CLK * !SN * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * QN) + (!CLK * !SN * QN)$	102.50600	102.50800	102.50800
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	46.31600	46.33000	46.96910

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(CLK * QN) + (!CLK * D * QN)$	0.00000	0.00000	0.00000
	$(CLK * QN) + (!CLK * D * QN)$	51.93220	51.93490	51.94570
	$(!CLK * !D * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !D * QN)$	39.59170	39.58620	40.16130

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(CLK * QN) + (!CLK * D * QN)$	0.00000	0.00000	0.00000
	$(CLK * QN) + (!CLK * D * QN)$	55.89430	55.90090	56.56230
	$(!CLK * !D * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !D * QN)$	11.14570	11.17040	12.16300

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(D * SN) + (D * !SN * QN)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * QN)$	56.75610	56.75840	57.43500
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	104.76600	104.76300	105.35500
	$(!D * !SN * QN)$	0.00000	0.00000	0.00000
	$(!D * !SN * QN)$	29.16720	29.18580	29.96280

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQNX1	$(D * SN) + (D * !SN * QN)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * QN)$	40.54460	40.54880	41.23620
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	53.35980	53.35710	53.98620
	$(!D * !SN * QN)$	0.00000	0.00000	0.00000
	$(!D * !SN * QN)$	10.12400	10.13710	11.02620

TMRDFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80,
Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNQX1	670.85199

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
TMRDFFSNQX1	0.04117	0.07156	0.07187	5.32224

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQX1	0.00000	85.26810	134.20100

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RR)	0.22598	0.33665	0.45991
	SN->Q (FR)	0.14525	0.30210	1.14402

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RF)	0.44969	0.53833	0.68157

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	0.01100	0.01196	0.51288
	setup	CLK (R)	0.12034	0.14686	1.04640

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	-0.07195	-0.12358	-1.15061
	setup	CLK (R)	0.09754	0.15110	1.49661

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	0.01100	0.01196	0.51288
	setup	CLK (R)	SN	0.12034	0.14686	1.04640

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	-0.07195	-0.12358	-1.15061
	setup	CLK (R)	SN	0.09754	0.15110	1.49661

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	0.02565	0.02663	3.91078
	removal	CLK (R)	-0.01681	-0.01495	-0.12028

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	!D	0.02565	0.02663	3.91078
	removal	CLK (R)	!D	-0.01681	-0.01495	-0.12028

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.09416	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.09416	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.10032	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.09724	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.14036	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15268	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.19581	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10032	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	59.18210	59.18100	59.84540
	SN	58.46730	58.46970	58.78090

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	105.56700	105.56700	106.22200

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	100.05800	100.06500	100.06700
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	40.16680	40.16930	40.77120

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	101.96200	101.96900	101.97200
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	54.53690	54.55270	55.20190

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	55.19270	55.18990	55.19960
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	44.14990	44.14560	44.73050

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	58.53340	58.53700	58.54070
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	12.03920	12.05230	12.67380

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	59.21830	59.22010	59.86870
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	105.39800	105.39600	106.03400
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	33.01260	33.03230	33.83200

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	40.88430	40.88670	41.58620
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	53.50330	53.49950	54.13780
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	11.54610	11.56000	12.43390

TMRDFFSNRNQNX1

sky130_rhbd_tt_1P8_25C.ccs

Cell Library: Process ,

Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
x	0	x	x	0
-	1	1	-	HiZ
x	1	0	x	1

Footprint

Cell Name	Area
TMRDFFSNRNQNX1	756.98798

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
TMRDFFSNRNQNX1	0.04229	0.10633	0.07714	0.07575	2.62885

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNRNQNX1	0.00000	95.56120	163.96400

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQNX1	CLK->Q (RR)	0.43152	0.53439	0.75093
	RN->Q (RR)	0.43074	0.55668	1.21725
	SN->Q (FR)	0.50123	0.61167	1.22702

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQNX1	CLK->Q (RF)	0.23680	0.32826	0.47727
	RN->Q (FF)	0.10878	0.26045	1.09617

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	-0.01181	-0.00969	1.96671
	setup	CLK (R)	0.15138	0.15744	0.63016

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	-0.06695	-0.11656	-1.03997
	setup	CLK (R)	0.10491	0.16144	2.17348

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.01181	-0.00969	1.96671
	setup	CLK (R)	(RN * SN)	0.15138	0.15744	0.63016

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	hold	CLK (R)	(RN * SN)	-0.06695	-0.11656	-1.03997
	setup	CLK (R)	(RN * SN)	0.10491	0.16144	2.17348

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	0.14171	0.15744	6.81466
	removal	CLK (R)	-0.01918	-0.02491	-0.06881
	hold	SN (R)	0.00778	-0.00747	-0.10458
	setup	SN (R)	0.06211	0.09965	0.86467

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	(D * SN)	0.14171	0.15744	6.81466
	removal	CLK (R)	(D * SN)	-0.01918	-0.02491	-0.06881
	hold	SN (R)	(CLK * D)	0.00778	-0.00747	-0.10458
	hold	SN (R)	(CLK * !D)	0.00628	-0.00996	-0.11214
	hold	SN (R)	(!CLK * D)	-0.04433	-0.07225	-0.22858
	hold	SN (R)	(!CLK * !D)	-0.04430	-0.07474	-0.22918
	setup	SN (R)	(CLK * D)	0.00036	0.01593	0.26715
	setup	SN (R)	(CLK * !D)	0.00187	0.01495	0.30103
	setup	SN (R)	(!CLK * D)	0.06211	0.09965	0.68167
	setup	SN (R)	(!CLK * !D)	0.06211	0.09965	0.86467

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.10956	0.40527	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10032	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.10340	0.40527	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.10340	0.40527	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	0.02600	0.01928	0.24554
	removal	CLK (R)	0.00517	0.00747	0.01523
	hold	RN (R)	0.06530	0.10961	0.50994
	setup	RN (R)	0.02498	0.01007	0.67769

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	recovery	CLK (R)	(!D * RN)	0.02600	0.01928	0.24554
	removal	CLK (R)	(!D * RN)	0.00517	0.00747	0.01523
	hold	RN (R)	CLK	0.02684	0.06228	0.36570
	hold	RN (R)	(!CLK * D)	0.06530	0.10463	0.48328
	hold	RN (R)	(!CLK * !D)	0.06279	0.10961	0.50994
	setup	RN (R)	CLK	0.02498	0.01007	0.67769
	setup	RN (R)	(!CLK * D)	-0.03105	-0.06379	0.56214
	setup	RN (R)	(!CLK * !D)	-0.03286	-0.06575	0.04934

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08800	0.40527	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08800	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.11880	0.40527	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.11880	0.40527	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.17733	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.18657	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.21737	0.40527	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11880	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	94.36770	94.37080	94.98120
	RN	95.18600	95.18850	96.30010
	SN	94.20900	94.23010	95.23880

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	97.41190	97.40880	98.00390
	RN	-0.00000	-0.00000	-0.00000
	RN	98.60640	98.63970	99.71630

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * RN * SN) + (CLK * RN * !SN * Q) + (CLK * !RN * !Q) + (!CLK * RN * !SN * Q) + (!CLK * !RN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN) + (CLK * RN * !SN * Q) + (CLK * !RN * !Q) + (!CLK * RN * !SN * Q) + (!CLK * !RN * !Q)$	97.65790	97.65830	97.84840
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	50.40040	50.40130	50.99360

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * RN * SN) + (CLK * RN * !SN * Q) + (CLK * !RN * !Q) + (!CLK * RN * !SN * Q) + (!CLK * !RN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN) + (CLK * RN * !SN * Q) + (CLK * !RN * !Q) + (!CLK * RN * !SN * Q) + (!CLK * !RN * !Q)$	95.19140	95.19190	95.19170
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	50.47100	50.48390	51.13090

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	91.36050	91.36140	91.35600
	$(!CLK * D * SN * !Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q)$	51.74780	51.74000	52.30750

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	96.64370	96.64580	96.64500
	$(!CLK * D * SN * !Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q)$	55.03020	55.03300	55.62850

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	88.03570	88.05040	88.05510
	$(!RN * !Q)$	0.00000	0.00000	0.00000
	$(!RN * !Q)$	98.10410	98.10800	99.29400
	$(!CLK * !D * RN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * RN * Q)$	49.41570	49.41120	49.98940

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	93.90290	93.91540	94.56360
	$(!RN * !Q)$	0.00000	0.00000	0.00000
	$(!RN * !Q)$	16.41480	16.43560	17.71840
	$(!CLK * !D * RN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * RN * Q)$	18.53580	18.56080	19.54360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(D * RN * SN) + (D * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * RN * SN) + (D * RN * !SN * Q)$	94.36550	94.36520	94.97110
	$(D * !RN * !Q) + (!D * RN * SN) + (!D * !RN * !Q)$	0.00000	0.00000	0.00000
	$(D * !RN * !Q) + (!D * RN * SN) + (!D * !RN * !Q)$	97.21810	97.21530	97.78520
	$(!D * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * RN * !SN * Q)$	51.20160	51.20580	51.96260

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQNX1	$(D * RN * SN) + (D * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * RN * SN) + (D * RN * !SN * Q)$	55.04190	55.04760	55.68320
	$(D * !RN * !Q) + (!D * RN * SN) + (!D * !RN * !Q)$	0.00000	0.00000	0.00000
	$(D * !RN * !Q) + (!D * RN * SN) + (!D * !RN * !Q)$	53.21190	53.22110	53.87210
	$(!D * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * RN * !SN * Q)$	16.45770	16.46740	17.32000

TMRDFFSNRNQX1

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*Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
x	1	1	x	IQ
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0

Footprint

Cell Name	Area
TMRDFFSNRNQX1	774.21521

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
TMRDFFSNRNQX1	0.04230	0.10517	0.07677	0.07390	5.43434

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNRNQX1	0.00000	103.31400	172.73700

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNRNQX1	CLK->Q (RR)	0.28450	0.38440	0.62472
	SN->Q (FR)	0.14815	0.30772	1.12074

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	-0.01357	-0.01593	2.24337
	setup	CLK (R)	0.15904	0.17309	0.63636

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	-0.06484	-0.11711	-1.09851
	setup	CLK (R)	0.12420	0.18485	3.32863

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.01357	-0.01593	2.24337
	setup	CLK (R)	(RN * SN)	0.15904	0.17309	0.63636

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06484	-0.11711	-1.09851
	setup	CLK (R)	(RN * SN)	0.12420	0.18485	3.32863

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	0.15136	0.18457	6.98764
	removal	CLK (R)	-0.01918	-0.02491	-0.06778
	hold	SN (R)	0.03273	0.05979	0.31254
	setup	SN (R)	0.01412	0.01593	1.17186

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.15136	0.18457	6.98764
	removal	CLK (R)	(D * SN)	-0.01918	-0.02491	-0.06778
	hold	SN (R)	(CLK * D)	0.00778	-0.00747	-0.09864
	hold	SN (R)	(CLK * !D)	0.00625	-0.00747	-0.10070
	hold	SN (R)	(!CLK * D)	0.03273	0.05979	0.29716
	hold	SN (R)	(!CLK * !D)	0.03273	0.05979	0.31254
	setup	SN (R)	(CLK * D)	0.00537	0.01593	0.52049
	setup	SN (R)	(CLK * !D)	0.00501	0.01593	0.58714
	setup	SN (R)	(!CLK * D)	0.01412	0.00604	1.17186
	setup	SN (R)	(!CLK * !D)	0.01276	0.00400	0.97659

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.19889	0.40527	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.22353	0.40527	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	102.82200	102.82000	103.41600
	SN	102.40900	102.42500	103.44400

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	96.68950	96.68630	97.07360
	$(CLK * RN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * RN * !SN * Q)$	102.76300	102.76000	103.34300
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	53.90450	53.90180	54.48760

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q) + (CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q) + (!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	95.15180	95.15110	95.15240
	$(CLK * RN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * RN * !SN * Q)$	68.00820	68.01550	68.43490
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	53.97230	53.98590	54.62500

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	91.12900	91.12700	91.12380
	$(CLK * D * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * D * !SN * Q)$	103.68300	103.68200	104.79600
	$(CLK * !D * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !D * !SN * Q)$	58.46890	58.46750	59.30570
	$(!CLK * D * SN * !Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q)$	52.00430	51.99830	52.56780
	$(!CLK * D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * !SN * Q)$	50.67180	50.66520	51.71240
	$(!CLK * !D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * !SN * Q)$	16.10690	16.10650	16.62130

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q) + (!CLK * !D * SN * !Q)$	98.89710	98.90180	99.58790
	$(CLK * D * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * D * !SN * Q)$	19.15880	19.19150	20.94780
	$(CLK * !D * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !D * !SN * Q)$	23.35280	23.37150	24.54230
	$(!CLK * D * SN * !Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q)$	56.18300	56.19660	57.10410
	$(!CLK * D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * !SN * Q)$	2.69829	2.72677	3.99835
	$(!CLK * !D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * !SN * Q)$	2.98429	2.99831	3.58933

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	97.05370	97.06050	97.06500
	$(!CLK * !D * RN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * RN * Q)$	58.34490	58.34280	58.92180

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * Q) + (!CLK * D * RN * Q)$	101.48200	101.48600	101.79300
	$(!CLK * !D * RN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * RN * Q)$	23.22130	23.24120	24.03670

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	$(D * RN * Q)$	0.00000	0.00000	0.00000
	$(D * RN * Q)$	102.74900	102.74600	103.33000
	$(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)$	97.70920	97.70620	98.31990
	$(!D * RN * SN * Q)$	0.00000	0.00000	0.00000
	$(!D * RN * SN * Q)$	97.82570	97.82890	98.43500
	$(!D * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * RN * !SN * Q)$	59.94480	59.95350	60.71660

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNRNQX1	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	51.42330	51.42850	52.05440
	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	53.93380	53.94070	54.58370
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * !RN * SN * !Q) + (D * !RN * !SN * Q) + (!D * RN * SN * !Q) + (!D * !RN * SN * !Q) + (!D * !RN * !SN * Q)	53.01920	53.03080	53.72010
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	58.21860	58.22620	58.95400
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	21.78840	21.80120	22.63150

VOTER3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	x	0
x	1	x	1
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
VOTER3X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
VOTER3X1	0.02195	0.02013	0.02058	5.47682

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTER3X1	0.00000	4.21851	10.36970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (RR)	0.07415	0.15153	-1.27740
	B->Y (RR)	0.07580	0.13471	-0.54005
	C->Y (RR)	0.06786	0.11542	-0.46580

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (FF)	0.12367	0.24925	1.36249
	B->Y (FF)	0.18818	0.35714	1.90007
	C->Y (FF)	0.15734	0.28273	1.72087

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	5.10330	5.10740	5.30580
	B	0.00000	0.00000	0.00000
	B	0.17467	0.17385	0.30046
	C	0.00000	0.00000	0.00000
	C	5.10094	5.10382	5.28807

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	-0.00000	-0.00000	-0.00000
	A	2.09334	2.09588	2.31216
	B	0.00000	0.00000	0.00000
	B	1.12891	1.12778	1.22872
	C	0.00000	0.00000	0.00000
	C	2.10423	2.10763	2.34500

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01494	0.01469	0.01467
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.11485	0.11276	0.11213
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	1.46009	1.45955	1.45937

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	1487.21000	1408.82000	1396.56000
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1865.56000	1842.45000	1835.99000
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22135	2.22077	2.22154

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02746	0.02438	0.02347
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	5.29956	5.36578	5.48567
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	3.15209	3.19463	3.28212

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	5.84802	5.84795	5.84803
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22399	2.22341	2.22334
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.21624	2.21454	2.21548

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.01750	0.01748	0.01752
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	6242.72000	6247.19000	6254.73000
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	6.25236	6.25485	6.25242
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	1.45648	1.45601	1.45520

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.16840	0.16843	0.16849
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	1778.88000	1776.86000	1773.80000
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	10.47090	9.12447	8.84120
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	2.22054	2.21910	2.21946

VOTERN3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	1	-	HiZ
1	1	x	1
x	0	x	1

Footprint

Cell Name	Area
VOTERN3X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
VOTERN3X1	0.02188	0.02022	0.02062	1.81204

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTERN3X1	0.00000	1.10578	10.79310

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (FR)	0.06788	0.16530	1.05741
	B->YN (FR)	0.12813	0.27732	1.55963
	C->YN (FR)	0.09725	0.20345	1.47910

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (RF)	0.03585	0.08039	-1.68215
	B->YN (RF)	0.03670	0.06418	-0.90729
	C->YN (RF)	0.02926	0.03567	-0.82876

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	0.00000	0.00000	0.00000
	A	0.05072	0.05286	0.24000
	B	0.06979	0.06788	0.14646
	C	0.00000	0.00000	0.00000
	C	0.06237	0.06477	0.27377

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	0.00000	0.00000	0.00000
	A	3.78792	3.78947	3.94401
	B	-0.00000	-0.00000	-0.00000
	B	0.18038	0.17770	0.26830
	C	0.00000	0.00000	0.00000
	C	3.78483	3.78671	3.93301

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.01177	0.01158	0.01153
	(B * !C * !YN)	0.00000	0.00000	0.00000
	(B * !C * !YN)	0.13720	0.13508	0.13440
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.00528	0.00524	0.00520

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.05435	0.05149	0.05087
	(B * !C * !YN)	0.00000	0.00000	0.00000
	(B * !C * !YN)	0.03803	0.03717	0.03653
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.02454	0.02482	0.02498

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.02677	0.02374	0.02281
	(!A * C * YN)	0.00000	0.00000	0.00000
	(!A * C * YN)	-0.00385	-0.00394	-0.00411
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	-0.00713	-0.00724	-0.00740

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	6.24536	6.24438	6.24404
	(!A * C * YN)	0.00000	0.00000	0.00000
	(!A * C * YN)	0.02811	0.02760	0.02754
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	0.02344	0.02293	0.02285

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	0.01459	0.01457	0.01458
	(!A * B * YN)	0.00000	0.00000	0.00000
	(!A * B * YN)	-0.00111	-0.00115	-0.00114
	(!A * B * !YN)	0.00000	0.00000	0.00000
	(!A * B * !YN)	0.01785	0.01777	0.01782
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	-0.00103	-0.00103	-0.00111

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	0.19822	0.19823	0.19831
	(!A * B * YN)	0.00000	0.00000	0.00000
	(!A * B * YN)	0.02401	0.02352	0.02347
	(!A * B * !YN)	0.00000	0.00000	0.00000
	(!A * B * !YN)	0.04034	0.04037	0.04049
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	0.02443	0.02391	0.02373

XNOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Footprint

Cell Name	Area
XNOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XNOR2X1	0.02194	0.02435	2.54427

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XNOR2X1	0.00000	9.81263	18.16860

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (RR)	B	0.05745	0.11675	-0.08237
	A->Y (FR)	!B	0.04725	0.14121	1.35642
	B->Y (RR)	A	0.07546	0.13492	-0.08466
	B->Y (FR)	!A	0.03793	0.13769	1.41719

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (FF)	B	0.05967	0.15841	0.76428
	A->Y (RF)	!B	0.03204	0.06377	-0.29507
	B->Y (FF)	A	0.07346	0.17685	0.79804
	B->Y (RF)	!A	0.02769	0.03715	-0.61440

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	4.99897	5.00524	5.24071
	A	!B	0.00000	0.00000	0.00000
	A	!B	0.07292	0.08886	0.65095
	B	A	0.00000	0.00000	0.00000
	B	A	1.83424	1.83585	1.96949
	B	!A	0.00000	0.00000	0.00000
	B	!A	0.06961	0.08857	0.70892

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	11.02950	11.03650	11.28020
	A	!B	0.00000	0.00000	0.00000
	A	!B	8.94461	8.95477	9.46706
	B	A	0.00000	0.00000	0.00000
	B	A	8.99283	8.99893	9.24290
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.34818	9.35196	9.75970

XOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Footprint

Cell Name	Area
XOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XOR2X1	0.02210	0.02440	2.57984

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XOR2X1	0.00000	9.77674	19.21570

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (RR)	!B	0.05731	0.11143	-0.17898
	A->Y (FR)	B	0.05014	0.13348	1.11423
	B->Y (RR)	!A	0.08181	0.13131	-0.15902
	B->Y (FR)	A	0.04176	0.10569	0.64024

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (FF)	!B	0.05931	0.16218	0.84805
	A->Y (RF)	B	0.02874	0.07116	-0.07151
	B->Y (FF)	!A	0.06673	0.17166	0.84016
	B->Y (RF)	A	0.02456	0.07413	0.02755

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	0.54064	0.55316	1.04971
	A	!B	0.00000	0.00000	0.00000
	A	!B	1.72961	1.73456	1.97536
	B	A	0.00000	0.00000	0.00000
	B	A	0.53082	0.54180	0.93207
	B	!A	0.00000	0.00000	0.00000
	B	!A	1.74644	1.74711	1.98918

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	15.38180	15.39840	16.05740
	A	!B	0.00000	0.00000	0.00000
	A	!B	6.22675	6.23464	6.47013
	B	A	0.00000	0.00000	0.00000
	B	A	15.38060	15.39770	16.07340
	B	!A	0.00000	0.00000	0.00000
	B	!A	6.23266	6.23980	6.44009